

Model Program Book



SHORT-TERM INTERNSHIP (On-Site/Virtual)

Designed & Developed by



**ANDHRA PRADESH
STATE COUNCIL OF HIGHER EDUCATION**

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PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

(Onsite / Virtual)

Name of the Student:

Name of the College:

Registration Number:

Period of Internship: From: To:

Name & Address of the Intern Organization

_____ **University**
YEAR

An Internship Report on AWS Academy Cloud Virtual Internship

Submitted in accordance with the requirement for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

submitted by

P.V.SasinKumar

(228A1A05C2)

Under the Guidance of

T.Srinivasulu

(Asst.professor)

CSE



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS
(INTEGRATED CAMPUS)**

(Approved by AICTE, NEW DELHI, Affiliated to JNTUK, KAKINADA)

VALLURU(Post), ONGOLE, PRAKASAM District, A.P- 523272.

2024-2025

RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS

Approved by AICTE, Permanently Affiliated to JNTUK & Accredited by NAAC with 'A' Grade

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the report entitled —**AWS ACADEMY CLOUD FOUNDATIONS**||, that is being submitted by **P.V.Sasi Kumar** of III Year I Semester bearing **228A1A05C2**, in partial fulfilment for the award of the Degree of Bachelor of Technology in **Computer Science and Engineering**, Rise Krishna Sai Prakasam Group Of Institutions is a record of bonafide work carried out by them.

Signature of Guide

Signature of Senior Faculty Member

Signature of Head of the Department

Signature of External Examiner

VISION AND MISSION OF THE INSTITUTION

Vision of the Institute	To be a premier institution in technical education by creating professionals of global standards with ethics and social responsibility for the development of the nation and the mankind.
Mission of the Institute	Impart Outcome Based Education through well qualified and dedicated faculty.
	Provide state-of-the-art infrastructure and facilities for application-oriented research.
	Reinforce technical skills with life skills and entrepreneurship skills.
	Promote cutting-edge technologies to produce industry-ready Professionals.
	Facilitate interaction with all stakeholders to foster ideas and innovation.
	Inculcate moral values, professional ethics and social responsibility

VISION AND MISSION OF THE DEPARTMENT

Vision of the Department	To be a center of excellence in computer science and engineering for value-based education to serve humanity and contribute for socio-economic development.
Mission of the Department	▪ Provide professional knowledge by student centric teaching-learning process to contribute software industry.
	▪ Inculcate training on cutting edge technologies for industry needs.
	▪ Create an academic ambiance leading to research.
	▪ Promote industry institute interaction for real time problem solving.

Program Outcomes (POs)

PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering Problems
PO2	Problem Analysis: Identify, formulate, review research literature, and analyze complex Engineering problem searching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and Environmental considerations.
PO4	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information provide valid conclusions
PO5	Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations
PO6	The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the Professional engineering practice.
PO7	Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the Knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice
PO9	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
PO11	Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments
PO12	Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Educational Objectives (PEOs):

PEO1:	Develop software solutions for real world problems by applying Mathematics
PEO2:	Function as members of multi-disciplinary teams and to communicate effectively using modern tools.
PEO3:	Pursue career in software industry or higher studies with continuous learning and apply professional knowledge.
PEO4:	Practice the profession with ethics, integrity, leadership and social responsibility.

Program Specific Outcomes (PSOs):

PSO1	Domain Knowledge: Apply the Knowledge of Programming Languages, Networks and Databases for design and development of Software Applications.
PSO2	Computing Paradigms: Understand the evolutionary changes in computing possess knowledge of context aware applicability of paradigms and meet the challenges of the future.

Student's Declaration

I, **P.V.Sasi Kumar** a student of BACHELOR OF TECHNOLOGY program, Reg.No: **228A1A05C2** of the Department of **COMPUTER SCIENCE AND ENGINEERING**, RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS do hereby declare that I have completed the mandatory internship from MAY 2024 to JUNE 2024 in AWS ACADEMY CLOUD FOUNDATION under the Faculty Guideship of **T.Srinivasulu** Department of COMPUTER SCIENCE AND ENGINEERING , RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS.

(Signature and Date)

COMPLETION CERTIFICATE



N·E·A·T

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Certificate of Virtual Internship

This is to certify that

Pancheti.Venkata Sasi Kumar

RISE Krishna Sai Prakasam Group of Institutions

has successfully completed 10 weeks

Cloud Virtual Internship

During April - June 2024

Curriculum Provided by:

aws academy

Shri Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
NEAT Cell, AICTE

Dr. Satya Ranjan Biswal
Chief Technology Officer (CTO)
EduSkills



Certificate ID :62810ec3f432593c523e322a12b4b30a

Student ID :STU661a3bb93fa2f1712995257



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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It gives us an immense Pleasure to express a deep sense of gratitude to our supervisor **T.Srinivasulu (Asst. Professor** Department of **COMPUTER SCIENCE AND ENGINEERING** because his whole hearted and invaluable guidance throughout the report. Without his sustained and sincere effort, this report would not have taken this shape. He encouraged and helped us to overcome various difficulties and that

We have faced at various stages of our report. We would like to sincerely thank **Dr. K.Narayana Rao, Professor & HOD**, Computer Science & Engineering , for providing all the necessary facilities that led to the successful completion of our report. We would like to take this opportunity to thank our beloved Principal **Dr. A. V, BHASKAR RAO, Ph. D, M. Tech**, for providing a great support to us in completing our project and for giving us the opportunity of doing the internship report.

Finally, we would like to thank all of our friends and family members for their continuous help and encouragement.

P.V.Sasi Kumar
228A1A05C2

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ACKNOWLEDGEMENT

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Course objectives and overview

To begin, it is important to have an understanding of the prerequisites for this course. First, you should have general IT technical knowledge. The foundational computer literacy skills you will need to be successful in this course include a knowledge of basic computer concepts, file management, and a good understanding of the internet. Second, you should have general IT business knowledge. This includes insight into how information technology is used by businesses and other organizations.

Additionally, to ensure success in this course, it is preferred that you have:

- A general familiarity with cloud computing concepts
- A working knowledge of distributed systems
- Familiarity with general networking concepts
- A working knowledge of multi

After completing this course, you should be able to:

- Define the AWS Cloud.
- Explain the AWS pricing philosophy.
- Identify the global infrastructure components of AWS
- Describe security and compliance measures of the AWS Cloud including AWS Identity and Access Management (IAM).
- Create an AWS Virtual Private Cloud (Amazon VPC).
- Demonstrate when to use Amazon Elastic Compute Cloud (EC2), AWS Lambda and AWS Elastic Beanstalk.
- Differentiate between Amazon S3, Amazon EBS, Amazon EFS and Amazon S3 Glacier.
- Demonstrate when to use AWS Database services including Amazon Relational Database Service (RDS), Amazon DynamoDB, Amazon Redshift, and Amazon Aurora.
- Explain AWS Cloud architectural principles.
- Explore key concepts related to Elastic Load Balancing (ELB), Amazon CloudWatch, and Auto Scaling.

Modules:

To achieve the course objectives, the course explores the following topics:

Module 1: Cloud Concepts Overview

Module 2: Cloud Economics and Billing

Module 3: AWS Global Infrastructure Overview

Module 4: AWS Cloud Security

Module 5: Networking and Content Delivery

Module 6: Compute

Module 7: Storage

Module 8: Databases

Module 9: Cloud Architecture

Module 10: Automatic Scaling and

MODULES

Module 1: Cloud Concepts Overview

Module sections:

- Introduction to cloud computing
- Advantages of cloud computing
- Introduction to Amazon Web Services (AWS)
- Moving to the AWS Cloud the AWS Cloud Adoption Framework (AWS CAF)



In this module, Section 1 introduces cloud computing. In Section 2, you learn about the advantages that cloud computing provides over a traditional, on-premises computing model. In Section 3, you learn about what AWS is and the broad range of AWS products and services. You become familiar with the idea that AWS services are designed to work together to build solutions that meet business goals and technology requirements. The module concludes with Section 4, which is about the AWS Cloud Adoption Framework (AWS CAF). It covers the fundamental changes that must be supported for an organization to successfully migrate its IT portfolio to the cloud.

Module 2: Cloud Economics and Billing

Module sections:

- Fundamentals of pricing
- Total Cost of Ownership
- AWS Organizations
- AWS Billing and Cost Management
- Technical support



The purpose of this module is to introduce you to the business advantages of moving to the cloud. Section 1 describes the principles for how AWS sets prices for the various services. This includes the AWS pricing model and a description of the AWS Free Tier:<https://aws.amazon.com/free> Section 2 describes the Total Cost of Ownership and how customers can reduce their costs by moving IT services to the cloud. The section outlines four types of costs that are reduced by using cloud computing, and provides examples that illustrate each of these type. Section3 describes how customers can use AWS Organizations to manage their costs. Section4 describes billing and the components of the AWS Billing dashboard. This section includes a demonstration of how customers can use the dashboard to understand and manage their cost Finally, Section 5 describes the four different options for AWS Technical Support: Basic Support, Developer Support, Business Support, and Enterprise Support. The section a includes an activity that will help you understand the benefits of each support option.

Module 3: AWS Global Infrastructure Overview

Module sections:

- AWS Global Infrastructure
- AWS services and service category overview



Module 3 provides an overview of the AWS global infrastructure. In Section 1, you are introduced to the major parts of the AWS Global Infrastructure, including Region Availability Zones, the network infrastructure, and Points of Presence. In Section 2, you are shown a listing of all the AWS service categories, and then you are provided with a listing of each of the services that this course will discuss. The module ends with an AWS Management Console clickthrough activity.

Module 4: AWS Cloud Security

Module sections:

- AWS shared responsibility model
- AWS Identity and Access Management (IAM)
- Securing a new AWS account
- Securing accounts
- Securing data on AWS
- Working to ensure compliance



This module provides an introduction to the AWS approach to security. In Section 1, you are introduced to the AWS shared responsibility model, specifies which responsibilities belong to the customer and which responsibilities belong to AWS. Section 2 introduces you to the key concepts of AWS Identity and Access Management (IAM), including users, groups, policies, and roles. Section 3 provides guidance on how to secure a new AWS account. It discusses how you should avoid using the AWS account root user for day-to-day activities. It also best practices, such as creating IAM users that have multi-factor authentication (MFA) enabled. Section 3 highlights other ways to secure accounts. It discusses the security-related features of AWS Organizations, which include service control policies. This section also discusses AWS Shield, Amazon Cognito, and AWS Key Management Service (AWS KMS). Section 5 discusses how to secure data on AWS. Topics include encryption of data at rest and data in transit, and discusses options for securing data that is stored on Amazon Simple Storage Service (Amazon S3).

Finally, Section 6 discusses how AWS supports customer efforts to deploy solutions that are in compliance with laws and regulations. It also discusses the certifications that AWS maintains and AWS services —such as AWS Config and AWS Artifact—that support compliance.

Module 5: Networking and Content Delivery

Module sections:

- Networking basics
- Amazon VPC
- VPC networking
- VPC security
- Amazon Route 53
- Amazon CloudFront



The purpose of this module is to introduce you to the fundamental of AWS networking And content delivery services: Amazon Virtual Private Cloud (Amazon VPC), Amazon Route 53, and Amazon CloudFront. You will have the opportunity to label a virtual private cloud (VPC) network architecture diagram, design a VPC, watch how a VPC is built, and finally build a VPC yourself. Section 1 discusses networking concepts that will be referenced throughout the rest of the module: network, subnet, IPv4 and IPv6 addresses, and Classless Inter-Domain Routing (CIDR) notation. Section 2 provides an overview of the key terminology and features of Amazon VPC, which you must be familiar with when you design and build your own virtual private clouds (VPCs). In Section 3, you learn about several important VPC networking options: internet gateway, network address translation (NAT) gateway, VPC endpoints, VPC sharing, VPC peering, AWS Site-to-Site VPN, AWS Direct Connect, and AWS Transit Gateway. In Section 4, you learn how to secure VPCs with network access control lists (networkACLs) and security groups. Section5 covers Domain Name System (DNS)resolution and Amazon Route 53. It also covers the topic of DNS failover, which introduces the topic of high availability that you will learn about in more detail in module 10.

Module 6: Compute

Module sections:

- Compute services overview
- Amazon EC2
- Amazon EC2 cost optimization
- Container services
- Introduction to AWS Lambda
- Introduction to AWS Elastic Beanstalk



This module provides an introduction to many of the compute services offered by AWS. Section 1 provides a high-level, compute services overview. Section 2 introduces you to the key concepts of Amazon Elastic Compute Cloud (Amazon EC2), including Amazon Machine Images (AMIs), instance types, network settings, user data scripts, storage options, security groups, key pairs, instance lifecycle phases, Elastic IP addresses, instance metadata, and the benefits of using Amazon CloudWatch for monitoring. Section 3 focuses on the four pillars of cost optimization, with an emphasis on cost optimization as it relates to Amazon EC2. Section 4 covers container services. It introduces Docker and the differences between virtual machines and containers. It then discusses Amazon Elastic Container Service (Amazon ECS), AWS Fargate, Kubernetes, Amazon Elastic Kubernetes Service (Amazon EKS), and Amazon Elastic Container Registry (Amazon ECR). Section 5 introduces serverless computing with AWS Lambda. Event sources and Lambda function configuration basics are introduced, and the section ends with examples of a schedule-based Lambda function and an event-based Lambda function. Finally, Section 6 describes the advantages of using AWS Elastic Beanstalk for web application deployments. It concludes with a hands-on activity where you deploy a simple web application to Elastic Beanstalk.

Module 7: Storage

Module sections:

- Amazon Elastic Block Store (Amazon EBS)
- Amazon Simple Storage Service (Amazon S3)
- Amazon Elastic File System (Amazon EFS)
- Amazon Simple Storage Service Glacier



Module 7 introduces you to the various options for storing data with AWS. The module provides an overview of storages services — which are based on four different storage technologies — so that you can choose a storage service for various use cases. Section 1 provides you with an overview of the functionality of Amazon Elastic Block Store (Amazon EBS) and a summary of common use cases. It also introduces the concept of block versus object storage, and how to interact with Amazon EBS through the AWS Management Console. Section 2 provides an overview of the functionality of Amazon Simple Storage Service (Amazon S3) and a summary of common use cases. It also describes how Amazon S3 scales as demand grows and discusses the concept of data redundancy. The section also contains a general overview of Amazon S3 pricing. Section 3 starts with an overview of the functionality of Amazon Elastic File Store (Amazon EFS) and a summary of common use cases. It also provides an overview of the Amazon EFS architecture and a list of common Amazon EFS resources. Finally, in Section 4, you are provided an overview of the functionality of Amazon Storage Service Glacier and a summary of common use cases. This last section also describes the lifecycle of migrating data from Amazon S3 to Amazon S3 Glacier.

Module 8: Databases

Module sections:

- Amazon Relational Database Service (Amazon RDS)
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora



This module introduces you to four of the most commonly used AWS database services, with an emphasis on differentiating which database service to select for various use cases. Section 1 provides an overview of the Amazon Relational Database Service (Amazon RDS). It describes the difference between a managed and unmanaged service, and provides an overview of how to provide a highly available Amazon RDS implementation. In Section 2, an overview of the Amazon DynamoDB services is provided. The section also describes how DynamoDB uses data partitioning to address scenarios that call for high data volumes and the ability to scale out on demand. Section 3 provides an overview of Amazon Redshift. The section describes the parallel processing architecture of Amazon Redshift, and how this architecture supports processing very large datasets. It also reviews some of the more common use cases for Amazon Redshift. Finally, Section 4 provides an overview of Amazon Aurora. The module describes the use cases where Amazon Aurora is a better solution than Amazon RDS. It also discusses how Amazon Aurora provides a more resilient database solution through the use of multiple Availability Zone.

Module 9: Cloud Architecture

Module sections:

- AWS Well - Architected Framework
- Reliability and availability
- AWS Trusted Advisor



The purpose of this module is to introduce you to designing and building cloud architectures according to best practices. In Section 1, you learn about the AWS Well - Architected Framework and its purpose, how the framework is organized, and its design principles and best practices. You will also learn how to use it to design a cloud architecture solution that is secure, performant, resilient, and efficient. Finally, this section also introduces the AWS Well - Architected Tool, which can be used to evaluate your architectural designs against AWS Well - Architected Framework best practices. In Section 2, you learn about reliability and high availability, which are two factors to consider when you design an architecture that can withstand failure. In Section 3, you learn about AWS Trusted Advisor. You can use this tool to evaluate and improve your AWS environment when you implement your architectural design.

Module 10: Automatic Scaling and Monitoring

Module sections:

- Elastic Load Balancing
- Amazon CloudWatch
- Amazon EC2 Auto Scaling



The purpose of this module is to introduce you to three fundamental AWS services that can be used together to build dynamic, scalable architectures. Section 1 introduces you to Elastic Load Balancing, which is a service that automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, IP addresses, and Lambda functions. Section 2 introduces you to Amazon CloudWatch, which is a service that provides you with Data and actionable insights to monitor your applications, respond to system - wide performance changes, optimize resource utilization, and get a unified view of operational health. Finally, Section 3 introduces you to the Amazon EC2 Auto Scaling features that help you maintain application availability and enable you to automatically add or remove EC2 instances according to conditions that you define.

AWS certification exam information

AWS certification exams



The AWS Certified Cloud Practitioner certification provides individuals in various and technology roles with a way to validate their AWS Cloud knowledge and professional credibility. This exam covers four domains, including cloud concepts, security, technology, and billing and pricing. The AWS Certified Cloud Practitioner exam is the only AWS certification exam that is classified as foundational (as shown on the previous slide). It is often the first AWS IT professionals attempt to obtain. Though this AWS Academy Cloud Foundations course is not listed in the AWS Certified Cloud Practitioner Exam Guide as one of the AWS options recommended to prepare for the exam, this course does cover many of the same topics that are covered by AWS commercial courses, such as AWS Technical Essentials, AWS Business Essentials, and AWS Cloud Practitioner Essentials. Therefore, the AWS Academy Cloud Foundations course you are taking now is a good way to help prepare yourself to take this exam. The services included in the AWS Certified Cloud Practitioner exam change as new services are added. At a minimum, you should be able to describe the overall functionality of a broad range of AWS services before taking the exam. For an overview of the AWS services see the Amazon Web Services Cloud Platform section of the Overview of Amazon Web Services whitepaper.

ACTIVITY LOG FOR THE FIRST WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
DAY 1 01/05/2024	Introduction to cloud computing	Learned about the introduction to cloud computing	
DAY 2 02/05/2024	Advantages of cloud computing	Learned about the advantages of cloud computing	
DAY 3 03/05/2024	Introduction to Amazon Web Services (AWS)	Learned about the introduction to amazon web services	
DAY 4 04/05/2024	Introduction to Amazon Web Services (AWS)	Learned about the introduction to amazon web services	
DAY 5 05/05/2024	AWS Cloud Adoption Framework (AWS CAF)	Learned about the aws cloud adoption framework	
DAY 6 06/05/2024	AWS Cloud Adoption Framework (AWS CAF)	Learned about the aws cloud adoption framework	

WEEKLY REPORT

WEEK-1 (From:01/05/2024 to 06/05/2024)

The objective of the activity done: The main objective of the first week activities is to know about the Introduction to cloud computing and how to complete certain modules as part of the internship.

Detailed Report:

Day-1:

Cloud computing is the on - demand delivery of compute power, database, storage, applications, and other IT resources via the internet with pay – as – you – go pricing. These resources run on server computers that are located in large data centers in different locations around the world. When you use a cloud service provider like AWS, that service provider owns the computers that you are using. These resources can be used together like building blocks to build solutions that help meet business goals and satisfy technology requirements. In the traditional computing model, infrastructure is thought of as hardware. Hardware solutions are physical, which means they require space, staff, physical security, planning, and capital expenditure. In addition to significant upfront investment, another prohibitive aspect of traditional computing is the long hardware procurement cycle that involves acquiring, provisioning, and maintaining on - premises infrastructure. With a hardware solution, you must ask if there is enough resource capacity or sufficient storage to meet your needs, and you provision capacity by guessing theoretical maximum peaks. If you don't meet your projected maximum peak, then you pay for expensive resources that stay idle. If you exceed your projected maximum peak, then you don't have sufficient capacity to meet your needs. And if your needs change, then you must spend the time, effort, and money required to implement a new solution.

Day-2:

Trade capital expense for variable expense:

Capital expenses (capex) are funds that a company uses to acquire, upgrade, and maintain physical assets such as property, industrial buildings, or equipment. Do you remember the data center example in the traditional computing model where you needed to rack and stack the hardware, and then manage it all? You must pay for everything in the data center whether you use it or not. By contrast, a variable expense is an expense that the person who bears the cost can easily alter or avoid. Instead of investing heavily in data centers and servers before you know how you will use them, you can pay only when you consume resources and pay only for the amount you consume. Thus, you save money on technology. It also enables you to adapt to new applications with as much space as you need in minutes, instead of weeks or days. Maintenance is reduced, so you can spend focus more on the core goals of your business.

Benefit from massive economies of scale:

By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay. You can deploy your application in multiple AWS Regions around the world with just a few clicks. As a result, you can provide a lower latency and better experience for your customers simply and at minimal cost.

Day-3:

In general, a web service is any piece of software that makes itself available over the internet or on private (intranet) networks. A web service uses a standardized format—such as Extensible Markup Language (XML) or JavaScript Object Notation (JSON)—for the request and the response of an application programming interface (API) interaction. It is not tied to any one operating system or programming language. It's describing via an interface definition file it is discoverable.

For example, say you're building a database application. Your customers might be sending data to your Amazon Elastic Compute Cloud (Amazon EC2) instances, which is a service in the compute category. These EC2 servers batch the data in one-minute increments and add an object per customer to Amazon Simple Storage Service (Amazon S3), the AWS storage service you've chosen to use. You can then use a nonrelational database like Amazon DynamoDB to power your application, for example, to build an index so that you can find all the objects for a given customer that were collected over a certain period. You might decide to run these services inside an Amazon Virtual Private Cloud (Amazon VPC), which is a service in the networking category. The purpose of this simple example is to illustrate that you can select web services from different categories and use them together to build a solution.

Day-4:

The array of AWS services can be intimidating as you start your journey into the cloud. This course focuses on some of the more common services in the following service categories: compute, storage, database, networking and content delivery, security, identity, and compliance, management and governance, and AWS cost management.

Legend:

- Amazon Elastic Block Store (Amazon EBS)
- Amazon Elastic Compute Cloud (Amazon EC2)
- Amazon Elastic Container Registry (Amazon ECR)
- Amazon Elastic Container Service (Amazon ECS)
- Amazon Elastic File System (Amazon EFS)
- Amazon Elastic Kubernetes Service (Amazon EKS)

- Amazon Relational Database Service (Amazon RDS)
- Amazon Simple Storage Service (Amazon S3)
- Amazon Virtual Private Cloud (Amazon VPC)
- AWS Identity and Access Management (IAM)
- AWS Key Management Service (AWS KMS)

AWS Management Console:

The console provides a rich graphical interface to a majority of the features offered by AWS. (Note: From time to time, new features might not have all of their capabilities included in the console when the feature initially launches.)

AWS Command Line Interface (AWS CLI):

The AWS CLI provides a suite of utilities that can be launched from a command script in Linux, macOS, or Microsoft Windows.

Software development kits (SDKs):

AWS provides packages that enable accessing AWS in a variety of popular programming languages. This makes it easy to use AWS in your existing applications and it also enables you to create applications that deploy and monitor complex systems entirely through code.

Day-5:

As you learned so far in this module, cloud computing offers many advantages over the traditional model. However, for most organizations, cloud adoption does not happen instantly. Technology is one thing, but an organization also consists of people and processes, and these three elements must all be in alignment for successful cloud adoption. Cloud computing introduces a significant shift in how technology is obtained, used, and managed. It also shifts organizations budget and pay for technology services. Cloud adoption requires that fundamental changes are discussed and considered across an entire organization. It also requires that stakeholders across all organizational units—both within and outside IT—support these new changes. In this last section, you learn about the AWS CAF, which was created to help organizations design and travel an accelerated path to successful cloud adoption.

Day-6:

Each organization's cloud adoption journey is unique. However, in order for any organization successfully migrate its IT portfolio to the cloud, three elements (that is, people, process, and technology) must be in alignment. Business and technology leaders in an organization must understand the organization's current state, target state, and the transition that is needed to achieve the target state so they can set goals and create processes for staff. The AWS Cloud Adoption Framework (AWS CAF) provides guidance and best practices

to help organizations identify gaps in skills and processes. It also helps organizations build a comprehensive approach to cloud computing—both across the organization and throughout the IT lifecycle—to accelerate successful cloud adoption. At the highest level, the AWS CAF organizes guidance into six areas of focus, called perspective. Perspectives span people, processes, and technology. Each perspective consists of a set of capabilities, which covers distinct responsibilities that are owned or managed by functionally related stakeholders. Capabilities within each perspective are used to identify which areas of an organization require attention. By identifying gaps, prescriptive work streams can be created that support a successful cloud journey.

ACTIVITY LOG FOR THE SECOND WEEK

Day & Date	Description of the daily activity	Learning Outcome	Person In-Charge Signature
DAY 1 07/05/2024	Fundamentals of pricing	Learn about the Fundamentals of pricing	
DAY 2 08/05/2024	Total Cost of Ownership	Learn about the Total Cost of Ownership	
DAY 3 09/05/2024	AWS Organizations	Learn about the AWS Organizations	
DAY 4 10/05/2024	AWS Billing and Cost Management	Learn about the AWS Billing and Cost Management	
DAY 5 11/05/2024	Technical Support	Learn about the Technical Support	
DAY 6 12/05/2024	Overview of the Billing Dashboard	Show the Overview of the Billing Dashboard	

WEEKLYREPORT

WEEK 2 (From:07/05/2024 to 12/05/2024)

Objective of the activity done: The main objective of second week activities is to know about flows TotalCost of Ownership, Technical Support. And how to complete given modules as a part of the internship.

Detailed Report:

Day-1:

There are three fundamental drivers of cost with AWS:

compute, storage, and outbound data transfer. These characteristics vary somewhat, depending on AWS product and pricing model you choose. In most cases, there is no charge for inbound data transfer or for data transfer between other AWS services within the same AWS Region. There are some exceptions, so be sure to verify data transfer rates before you begin to use the AWS service. Outbound data transfer is aggregated across services and then charged at the outbound data transfer rate. This charge appears on the monthly statement as AWS Data Transfer Out.

Day-2:

On-premises versus cloud is a question that many businesses ask. The difference between these two options is how they are deployed. An on - premises infrastructure is installed locally on a company's own computers and servers. There are several fixed costs, also known as capital expenses, that are associated with the traditional infrastructure. Capital expenses include facilities, hardware, licenses, and maintenance staff. Scaling up can be expensive and time-consuming. Scaling down does not reduce fixed costs.

A cloud infrastructure is purchased from a service provider who builds and maintains the facilities, hardware, and maintenance staff. A customer pays for what is used. Scaling up or down is simple. Costs are easy to estimate because they depend on service use. It is difficult to compare premises IT delivery model with the AWS Cloud. The two are different because they use different concepts and terms. Using on premises IT involves a discussion that is based on capital expenditure, long planning cycles, and multiple components to buy, build, manage, and refresh resources over time. Using the AWS Cloud involves a discussion about flexibility, agility, and consumption-based costs.

Day-3:

AWS Organizations is a free account management service that enables you to consolidate multiple AWS accounts into an organization that you create and centrally manage. AWS Organizations include consolidated billing and account management capabilities that help you to better meet the budgetary, security, and compliance needs of your business.

The main benefits of AWS Organizations are:

Centrally managed access policies across multiple AWS accounts.

Controlled access to AWS services.

Automated AWS account creation and management.

Consolidated billing across multiple AWS accounts.

Here is some terminology to understand the structure of AWS Organizations. The diagram shows a basic organization, or root, that consists of seven accounts that are organized into four organizational units (or OUs). An OU is a container for accounts within a root. An OU can also contain other OUs. This structure enables you to create a hierarchy that looks like an upside-down tree with the root at the top. The branches consist of child OUs and they move downward until they end in accounts, which are like the leaves of the tree.

When you attach a policy to one of the nodes in the hierarchy, it flows down and it affects all the branches and leaves. This example organization has several policies that are attached to some of the OUs or are attached directly to accounts. An OU can have only one parent and, currently, each account can be a member of exactly one OU. An account is a standard AWS account that contains your AWS resources. You can attach a policy to an account to apply controls to only that one.

Day-4:

AWS Billing and Cost Management is the service that you use to pay your AWS bill, monitor usage, and budget your costs. Billing and Cost Management enables you to forecast and obtain a better idea of what your costs and usage might be in the future so that you can plan ahead. You can set a custom time period and determine whether you would like to view your data at a monthly or daily level of granularity. With the filtering and grouping functionality, you can further analyze your data using a variety of Available dimensions. The AWS Cost and Usage Report Tool enables you to identify the opportunities for optimization by understanding your cost and usage data trends and how you are using your AWS implementation. The AWS Billing and Cost Management console that includes the Cost Explorer page for viewing your AWS cost data as a graph. With Cost Explorer, you can visualize, understand, and manage your AWS costs and usage over time. The Cost Explorer includes a default report that visualizes your costs and usage for your top cost incurring AWS services. The monthly running costs report gives you an overview of all your cost for the past 3 months. It also provides forecasted numbers for the coming month, with a corresponding confidence interval. The Cost Explorer is a free tool that enables you to:

- View charts of your costs.
- View cost data for the past 13 months.
- Forecast how much you are likely to spend over the next 3 months.
- Discover patterns in how much you spend on AWS resources over time and identify cost problem areas.
- Identify the services that you use the most view metrics, like which Availability Zones have the most traffic or which linked AWS account is used the most.

Day-5:

AWS Budgets uses the cost visualization that is provided by Cost Explorer to show you the status of your budgets and to provide forecasts of your estimated costs. you can also use AWS Budgets to create notifications for when you go over your budget for the month, or when your estimated costs exceed your budget. Budgets can be tracked at the monthly, quarterly, or yearly level, and you can customize the start and end dates. Budget alerts can be sent via email or via Amazon Simple Notification Service (Amazon SNS). for accessibility:

The AWS Billing budgets panel showing budget names, current and future costs and usages, and headings for current and forecasted versus budgets. End of accessibility description. The AWS Cost and Usage Report is a single location for accessing comprehensive information about your AWS costs and usage. This tool lists the usage for each service to category that is used by an account (and its users) in hourly or daily line items, and any tax that you activated for tax allocation purposes. You can choose to have AWS to publish billing reports to an S3 bucket. These reports can be updated once a day. Whether you are new or continuing to adopt AWS services and applications as your business solutions, AWS want help you do amazing things with AWS. AWS Support can provide you with a unique combination of tools and expertise based on your current or future and planned use cases. AWS Support was developed to provide complete support and the right resources to aid your success. We want to support all our customers, including customers that might be experimenting with AWS, those that are looking for production uses of AWS, and also customers that use AWS as a business-critical resource. AWS Support can vary the type of support that is provided, depending on the customer's needs and goals.

Day-6:

In summary you:

- Explored the fundamentals of AWS pricing
- Reviewed Total Cost of Ownership concepts
- Reviewed an AWS Pricing Calculator estimate.

Total Cost of Ownership is a concept to help you understand and compare the costs that are associated with different deployments. AWS provides the AWS Pricing Calculator to assist you with the calculations that are needed to estimate cost savings.

Use the AWS Pricing Calculator to:

- Estimate monthly costs
- Identify opportunities to reduce monthly costs
- Model your solutions before building them
- Explore price points and calculations behind your estimate
- Find the available instance types and contracts that meet your needs

AWS Billing and Cost Management provides you with tools to help you access, understand, allocate, control, and optimize your AWS costs and usage. These tools include AWS Bills, AWS Cost Explorer, AWS

Budgets, and AWS Cost and Usage Reports. information about your AWS costs and usage including which AWS services are the main cost drivers.

ACTIVITY LOG FOR THE THIRD WEEK

Day & Date	Description of the daily activity	Learning Outcome	Person In Charge Signature
DAY 1 13/05/2024	LiveSession-5Content– Lightning Web Components (LWC)	Lightning Web Components (LWC)	
DAY 2 14/05/2024	Self-Paced Learning ModulesVS Code Setup	VS Code Setup installation, some concepts.	
DAY 3 15/05/2024	Self-Paced Learning ModulesCLC Setup	VS Code Setup installation, some concepts.	
DAY 4 16/05/2024	LiveSession-6Content: LightningWeb Components (LWC & API)	Basics Of LWC, Visual force, Use Lightning, Components in Lightning Experience, REST API, SOAPAPI	
DAY 5 17/05/2024	Self-Paced Learning ModulesAPI Basics Event Monitoring	Learned the basics of API, put the web in web API, Download and visualize Event Log files.	
DAY 6 18/05/2024	Self-Paced Learning Modules Shield Platform Encryption Apex Integration Services	Setup and Manage shield platform Encryption, Deploy shield platform encryption in a smart way, Apex SOAP and REST Callouts.	

WEEKLYREPORT

WEEK-3 (From:13/05/2024 to 18/05/2024)

The objective of the activity done: The main objective of the third week activities to

know about the Lightning Web Components (LWC), Lightning Web Components (LWC & API).

Detailed Report:

Day-1:

The AWS Cloud infrastructure is built around Regions. AWS has 22 Regions worldwide. An AWS Region is a physical geographical location with one or more Availability Zones. Availability Zones in turn consist of one or more data centers. To achieve fault tolerance and stability, Regions are isolated from one another. Resources in one Region are not automatically replicated to other Regions. When you store data in a specific Region, it is not replicated outside that Region. It is your responsibility to replicate data across Regions, if your business needs require it. AWS Regions that were introduced before March 20, 2019 are enabled by default. Regions that were introduced after March 20, 2019—such as Asia Pacific (Hong Kong) and Middle East (Bahrain)—are disabled by default. You must enable these Regions before you can use them. You can use the AWS Management Console to enable or disable a Region. Some Regions have restricted access. An Amazon AWS (China) account provides access to the Beijing and Ningxia Regions only. To learn more about AWS in China, see: <https://www.amazonaws.cn/en/about-aws/china/>. The isolated AWS GovCloud (US) Region is designed to allow US government agencies and customers to move sensitive workloads into the cloud by addressing their specific regulatory compliance requirements.

For accessibility:

Snapshot from the infrastructure. Aws website that shows a picture of downtown London including the Tower Bridge and the Shard. It notes that there are three Availability Zones in the London region. End of accessibility description.

Day-2:

As discussed previously, the AWS Global Infrastructure can be broken down into three elements: Regions, Availability Zones, and Points of Presence, which include edge locations. This infrastructure provides the platform for a broad set of services, such as networking, storage, compute services, and databases—and these services are delivered as an on-demand utility that is available in seconds, with pay-as-you-go pricing.

For accessibility:

Marketing diagram showing infrastructure at the bottom, consisting of Regions, Availability Zones, and edge locations. The next level up is labeled Foundational Services and includes graphics for compute, networking, and storage. That level is highlighted. Next level up platform services that includes databases, analytics, app services, deployment and management, and mobile services. Top layer is labeled applications and includes virtual desktops and collaboration and sharing. End of accessibility description.

Day-3:

AWS offers a broad set of cloudbased services. There are 23 different product or service categories, and each category consists of one or more services. This course will not attempt to introduce you to each service. Rather, the focus of this course is on the services that are most widely used and offer the best introduction to the AWS Cloud. This course also focuses on services that are more likely to be covered in the AWS Certified Cloud Practitioner exam. The categories that this course will discuss are highlighted on the slide: Compute, Cost Management, Database, Management and Governance, Networking and Content Delivery, Security, Identity, and Compliance, and Storage.

Day-4:

If you click Amazon EC2, it takes you to the Amazon EC2 page. Each product page provides a detailed description of the product and lists some of its benefits. Explore the different service groups to understand the categories and services within them. Now that you know how to locate information about different services, this module will discuss the highlighted service categories. The next seven slides list the individual services—within each of the categories and highlighted above—that this course will discuss.

Day-5:

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers scalability, data availability, security, and performance. Use it to store and protect any amount of data for websites, mobile apps, backup and restore, archive, enterprise applications, Internet of Things (IoT) devices, and big data analytics. Amazon Elastic Block Store (Amazon EBS) is high-performance block storage that is designed Use with Amazon EC2 for both throughput and transaction intensive workloads. It is used for a broad range of workloads, such as relational and non- relational databases, enterprise applications, containerized applications, big data analytics engines, file systems, and media workflows. Amazon Elastic File System (Amazon EFS) provides a scalable, fully managed elastic Network File System (NFS) file system for use with AWS Cloud services and on-premises of resources. It is built to scale on demand to petabytes, growing and shrinking automatically as you add and remove files. It reduces the need to provision and manage capacity to accommodate growth. Amazon Simple Storage Service Glacier is a secure, durable, and extremely low - cost Amazon S3 cloud storage class for data archiving and long-term backup. It is designed to deliver 11 9s of durability, and to provide comprehensive security and the capabilities to meet stringent regulatory requirements.

Day-6:

AWS compute services include the services listed here, and many others. Amazon Elastic Compute Cloud (Amazon EC2) provides resizable compute capacity as virtual machines in the cloud. Amazon EC2 Auto Scaling enables you to automatically add or remove EC2 instances according to conditions that you define. Amazon Elastic Container Service (Amazon ECS) is a highly scalable, high -performance container

orchestration service that supports Docker containers. Amazon Elastic Container Registry (Amazon ECR) is a fully- managed Docker container registry that makes it easy for developers to store, manage, and deploy Docker container images. AWS Elastic Beanstalk is a service for deploying and scaling web applications and services on familiar servers such as Apache and Microsoft Internet Information Services (IIS). AWS Lambda enables you to run code without provisioning or managing servers. you pay only for the compute time that you consume. There is no charge when your code is not running Amazon Elastic Kubernetes Service (Amazon EKS) makes it easy to deploy, manage, and scale containerized applications that use Kubernetes on AWS. AWS Fargate is a compute engine for Amazon ECS that allows you to run containers without having to manage servers or cluster.

ACTIVITY LOG FOR THE FOURTH WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In -Charge Signature
DAY 1 19/05/2024	AWS shared responsibility model	Learned the aws global infrastructure	
DAY 2 20/05/2024	AWS Identity and Access Management (IAM)	Learned aws services and service category overview	
DAY 3 21/05/2024	Securing a new AWS account	Learned aws management console click through	
DAY 4 22/05/2024	Securing accounts	Learned aws management console click through.	
DAY 5 23/05/2024	Securing data on AWS	Learned aws global infrastructure	
DAY 6 24/05/2024	Working to ensure compliance	Learned aws global infrastructure	

WEEKLYREPORT

WEEK 4 (From 19/05/2024 to 24/05/2024)

The objective of the activity done: The main objective of the fourth week is to test with in the knowledge gain all over the course by completing AWS Identity and the Management (IAM), Securing a new AWS account Securing a new AWS account, Securing accounts e.t.c

Detailed Report:

Day1:

Security and compliance are a shared responsibility between AWS and the customer. This share responsibility model is designed to help relieve the customer's operational burden. At the same time, to provide the flexibility and customer control that enables the deployment of customer solutions on AWS, the customer remains responsible for some aspects of the overall security. The differentiation of who is responsible for what is commonly referred to as security —off the cloud versus security —in the cloud. AWS operates, manages, and controls the components from the software virtualization layer down to the physical security of the facilities where AWS services operate. AWS is responsible protecting infrastructure that runs all the services that are offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run the AWS Cloud services. The customer is responsible for the encryption of data at rest and data in transit. The customer should also ensure that the network is configured for security and that security credentials and logins are managed safely. Additionally, the customer is responsible for the configuration of security groups and the configuration of the operating system that run on compute instances that they launch (including updates and security patches).

Day2:

AWS Identity and Access Management (IAM) allows you to control access to compute, storage, database, and application services in the AWS Cloud. IAM can be used to handle authentication, and to specify and enforce authorization policies so that you can specify which users can access services. IAM is a tool that centrally manages access to launching, configuring, managing, and terminating resources in your AWS account. It provides granular control over access to resources, including the ability to specify exactly which API calls the user is authorized to make to each service. Whether you use the AWS Management Console, the AWS CLI, or the AWS software development kits (SDKs), every call to an AWS service is an API call. With IAM, you can manage which resources can be accessed by who, and how these resources can be accessed. You can grant different permissions to different people for different resources. For example, you might allow some users full access to Amazon EC2, Amazon S3, Amazon DynamoDB, Amazon Redshift, and other AWS services. However, for other users, you might allow read-only access to only a few S3 buckets. Similarly, you might grant permission to other users to administer only specific EC2 instances.

Day3:

When you first create an AWS account, you begin with a single sign-in identity that has complete access to all AWS services and resources in the account. This identity is called the AWS account root user and it is accessed by signing into the AWS Management Console the email address and password that you used to create the account. AWS account root users have (and retain) full access to all resources in the account. Therefore, AWS strongly recommends that you do not use account root user credentials for day-to-day interactions with the account. Instead, AWS recommends that you use IAM to create additional users and assign permissions to these users, following the principle of least privilege. For example, if you require administrator-level permissions, you can create an IAM user, grant that user full access, and then use those credentials to interact with the account. Later, if you need to revoke or modify your permissions, you can delete or modify any policies that are associated with that IAM user. Additionally, if you have multiple users that require access to the account, you can create unique credentials for each user and define which user will have access to which resources. For example, you can create IAM users with read-only access to resources in your AWS account and distribute those credentials to users that require read access. You should avoid sharing the same credentials with multiple users. While the account root user should not be used for routine tasks, there are a few tasks that can only be accomplished by logging in as the account root user. A full list of these tasks is detailed on the [Tasks that require root user credentials AWS documentation page](#).

Day4:

AWS Organizations is an account management service that enables you to consolidate multiple AWS accounts into an organization that you create and centrally manage. Here, the focus is on the security features that AWS Organizations provides. One helpful security feature is that you can group accounts into organizational units (OUs) and attach different access policies to each OU. For example, if you have accounts that should only be allowed to access the AWS services that meet certain regulatory requirements, you can put those accounts into one OU. You then can define a policy that blocks OU access to services that do not meet those regulatory requirements, and then attach the policy to the OU. Another security feature is that AWS Organizations integrates with and supports IAM. AWS Organizations expands that control to the account level by giving you control over what users and roles in an account or a group of accounts can do. The resulting permissions are the logical intersection of what is allowed by the AWS Organizations policy settings and what permissions are explicitly granted by IAM in the account for that user or role. The user can access only what is allowed by both the AWS Organizations policies and IAM policies. Finally, AWS Organizations provides service control policies (SCPs) that enable you to specify the maximum permissions that member accounts in the organization can have. In SCPs, you can restrict which AWS services, resources, and individual actions the users and roles in each member account can access. These restrictions even override the administrators of

member accounts. When AWS Organizations blocks access to a service, resource, or API action, a user or role in that account can't access it, even if an administrator of a member account explicitly grants such permissions.

Day5:

Data encryption is an essential tool to use when your objective is to protect digital data. Data encryption takes data that is legible and encodes it so that it is unreadable to anyone who does not have access to the secret key that can be used to decode it. Thus, even if an attacker gains access to your data, they cannot make sense of it. Data at rest refers to data that is physically stored on disk or on tape. You can create encrypted file systems on AWS so that all your data and metadata is encrypted at rest by using the open standard Advanced Encryption Standard (AES)-256 encryption algorithm. When you use AWS KMS, encryption and decryption are handled automatically transparently, so that you do not need to modify your applications. If your organization is subject to corporate or regulatory policies that require encryption of data and metadata at rest, AWS recommends enabling encryption on all services that store your data. You can encrypt data stored in any service that is supported by AWS KMS. See [How AWS Services use AWS KMS](#) for a list of supported services.

Day6:

As an example of a certification for which you can use AWS services to meet your compliance goals, consider the ISO/IEC 27001:2013 certification. It specifies the requirements for establishing, implementing, maintaining, and continually improving an Information Security Management System. The basis of this certification is the development and implementation of a rigorous security program, which includes the development and implementation of an Information Security Management System. The Information Security Management System defines how AWS perpetually manages security in a holistic, comprehensive manner. AWS also provides security features and legal agreements that are designed to help support customers with common regulations and laws. One example is the Health Insurance Portability and Accountability Act (HIPAA) regulation. Another example, the European Union (EU) General Data Protection Regulation (GDPR) protects European Union data subjects' fundamental right to privacy and the protection of personal data. It introduces robust requirements that will raise and harmonize standards for data protection, security, and compliance. AWS Config is a Regional service. To track resources across Regions, enable it in every Region that you use. AWS Config offers an aggregator feature that can show an aggregated view of resources across multiple Regions and even multiple accounts.

ACTIVITY LOG FOR THE FIFTH WEEK

Day & Date	Description of the daily activity	Learning Outcome	Person Incharge Signature
DAY 1 25/05/2024	Networking basics	Learn about the Networking basics	
DAY 2 26/05/2024	Amazon VPC	Learn about the Amazon VPC.	
DAY 3 28/05/2024	VPC networking	Learned about VPC networking	
DAY 4 30/05/2024	VPC security	Learn about the VPC security	
DAY 5 31/05/2024	Amazon Route 53	Learned and create the Amazon Route 53	
DAY 6 01/06/2024	Amazon CloudFront	Learn the Amazon cloudfront	

WEEKLYREPORT

WEEK-5 (From 25/05/2024 to 01/06/2024)

The objective of the activity done: The main objective of this week activities is to know about the Validations and Networking basics, Amazon VPC e.t.c based on certain modules as part of the internship.

Detailed Report:

Day-1:

A computer network is two or more client machines that are connected together to share resources. A network can be logically partitioned into subnets. Networking requires a networking device (such as a router or switch) to connect all the clients together and enable communication between them.

Day-2:

Many of the concepts of an on-premises network apply to a cloud-based network, but much of the complexity of setting up a network has been abstracted without sacrificing control, security, and usability. In this section, you learn about Amazon VPC and the fundamental components of a VPC.

Day-3:

Now that you have learned about the basic components of a VPC, you can start routing traffic in interesting ways. In this section, you learn about different networking options.

Day-4:

You can build security into your VPC architecture in several ways so that you have complete control over both incoming and outgoing traffic. In this section, you learn about two Amazon VPC firewall options that you can use to secure your VPC: security groups and network access control lists (network ACLs).

Day-5:

Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service. It is designed to give developers and businesses a reliable and cost-effective way to route users to internet applications by translating names (like `www.example.com`) into the numeric IP addresses (like `192.0.2.1`) that computers use to connect to each other.

Day-6:

The purpose of networking is to share information between connected resources. So far in this module, you learned about VPC networking with Amazon VPC. You learned about the different options for connecting your VPC to the internet, to remote networks, to other VPCs, and to AWS services. Content delivery occurs over networks, too—for example, when you stream a movie from your favorite streaming service. In this final section, you learn about Amazon CloudFront, which is a content delivery network (CDN) service.

ACTIVITY LOG FOR THE SIXTH WEEK

Day & Date	Brief Description of the daily activity	Learning Outcome	Person Incharge Signature
DAY 1 02/06/2024	Compute services overview	To compute services overview	
DAY 2 03/06/2024	Amazon EC2	Amazon EC2	
DAY 3 04/06/2024	Amazon EC2 cost optimization	Learn about the Amazon EC2 cost optimization	
DAY 4 05/06/2024	Container services	Learn about the Container services	
DAY 5 06/06/2024	Introduction to AWS Lambda	Learn about the introduction to AWS lambda	
DAY 6 07/06/2024	Introduction to AWS Elastic Beanstalk	Learn about the introduction to AWS elastic beanstalk	

WEEKLY REPORT

WEEK-6 (From 02/06/2024 to 07/06/2024)

The objective of the activity done: The main objective of this week activities is to

Clarify the doubts and increasing the soft skills.

Detailed Report:

Day-1:

Section 2 includes a recorded Amazon EC2 demonstration. The end of this same section includes a hands-on lab, where you will practice launching an EC2 instance by using the AWS Management Console. There is also an activity in this section that has you compare advantages and disadvantages of running a database deployment on Amazon EC2, running it on Amazon Relational Database Service (RDS). Section 5 includes a hands-on AWS Lambda activity and section 6 includes a hands-on Elastic Beanstalk activity. Finally, you will be asked to complete a knowledge check that will test your understanding of the key concepts that are covered in this module.

Day-2:

Running servers on-premises is an expensive undertaking. Hardware must be procured, and this procurement can be based on project plans instead of the reality of how the servers are used. Data centers are expensive to build, staff, and maintain. Organizations also need to permanently provision a sufficient amount of hardware to handle traffic spikes and peak workloads. After traditional on-premises deployments are built, server capacity might be unused and idle for a significant portion of the time that the servers are running, which is wasteful.

Day-3:

The demonstration shows:

- How to use the AWS Management Console to launch an Amazon EC2 instance (with all the default instance settings accepted).
- How to connect to the Windows instance by using a Remote Desktop client and the key pair that was identified during instance launch to decrypt the Windows password for login.
- How to terminate the instance after it is no longer needed

Day-4:

The objective of this activity is to demonstrate that you understand the differences between building a deployment that uses Amazon EC2 and using a fully managed service, such as the Amazon RDS, to deploy your solution. At the end of this activity, you should be prepared to discuss the advantages and disadvantages of deploying Microsoft SQL Server on Amazon EC2 versus deploying it on Amazon RDS.

Day-5:

As you saw in the earlier sections of this module, AWS offers many compute options. For some examples, Amazon EC2 provides virtual machines. As another example, Amazon ECS and Amazon EKS are container-based compute services. However, there is another approach to compute that does not require you to provision or manage servers. This third approach is often referred to as serverless computing.

Day-6:

AWS Elastic Beanstalk is another AWS compute service option. It is a platform as a service (or PaaS) that facilitates the quick deployment, scaling, and management of your web and the applications and services. you remain in control. The entire platform is already built, and you only need to upload your code. Choose your instance type, your database, set and adjust the automatic scaling, update your application, access the server log files, and enable HTTPS on the load balancer.

ACTIVITY LOG FOR THE SEVEN AND EIGHTWEEK

Day & Date	Brief Description of the daily activity	Learning Outcome	Person Incharge Signature
DAY 1 08/06/2024	Amazon Elastic Block Store (Amazon EBS)	Amazon Elastic Block Store	
DAY 2 09/06/2024	Amazon Simple Storage Service Glacier	Amazon Simple Storage Service Glacier	
DAY 3 10/06/2024	Amazon Relational Database Service (Amazon RDS)	Amazon Relational Database Service	
DAY 4 11/06/2024	Amazon DynamoDB	Amazon DynamoDB	
DAY 5 12/06/2024	Amazon Redshift	Amazon Redshift	
DAY 6 13/06/2024	Amazon Aurora	Amazon Aurora	

WEEKLY REPORT

WEEK-7&8 (From 08/06/2024 to 13/06/2024)

The objective of the activity done: The main objective of this week activities is to Clarify the doubts and increasing the soft skills.

Detailed Report:

Day-1:

Amazon EBS provides persistent block storage volumes for use with Amazon EC2 instances. It Persistent storage is any data storage device that retains data after power to that device is shut off. It is also sometimes called non-volatile storage. Each Amazon EBS volume is only the automatically replicated within its Availability Zone to protect you from component failure. It is designed for high availability and durability. Amazon EBS volumes provide consistent and performance that is needed to run your workloads. With Amazon EBS, you can scale your usage up or down within minutes, while paying a low price for only what you provision.

Day-2:

The demonstration shows how to configure the following resources by using the AWS Management Console. The demonstration shows how to:

- Create an Amazon Glacier vault.
- Upload archived items to the vault using a third-party graphical interface tool.

In this educator- led activity, you will be asked to log in to the AWS Management Console. The activity instructions are on the next slide. You will be challenged to answer five questions. The educator will lead the class in a discussion of each question, and reveal the correct answers.

Day-3:

AWS solutions typically fall into one of two categories: unmanaged or managed. Unmanaged services are typically provisioned in discrete portions as specified by the user. You must manage how the service responds to changes in load, errors, and situations where resources become unavailable. Say that you launch a web server on an Amazon Elastic Compute Cloud (Amazon EC2) instance. Because Amazon EC2 is an unmanaged solution, that web server will not scale to handle increased traffic load or replace unhealthy instances with healthy ones unless you specify that it use a scaling solution, such as AWS Automatic Scaling. The benefit to using an unmanaged service is that you have more fine-tuned control over how your solution handles changes in load, errors, and situations where resources become unavailable.

Day-4:

AWS solutions typically fall into one of two categories: unmanaged or managed. Unmanaged services are typically provisioned in discrete portions as specified by the user. You must manage how the service responds

to changes in load, errors, and situations where resources become unavailable. Say that you launch a web server on an Amazon Elastic Compute Cloud (Amazon EC2) instance. Because Amazon EC2 is an unmanaged solution, that web server will not scale to handle increased traffic load or replace unhealthy instances with healthy ones unless you specify that it use a scaling solution, such as AWS Automatic Scaling. The benefit to using an unmanaged service is that you have more finetuned control over how your solution handles changes in load, errors, and situations where resources become unavailable. Amazon manages all the underlying data infrastructure for this service and redundantly stores data across multiple facilities in a native US Region as part of the fault-tolerant architecture. With DynamoDB, you can create tables and items. You can add items to a table. The system automatically partitions your data and has table storage to meet workload requirements. There is no practical limit on the number of items that you can store in a table. For instance, some customers have production tables that contain billions of items.

Day-5:

Analytics is important for businesses today, but building a data warehouse is complex and expensive. Data warehouses can take months and significant financial resources to set up. Amazon Redshift is a fast and powerful, fully managed data warehouse that is simple and cost-effective to set up, use, and scale. It enables you to run complex analytic queries against petabytes of structured data by using sophisticated query optimization, columnar storage on high-performance local disks, and massively parallel data processing. Most results come back in seconds.

Day-6:

Amazon Aurora is a MySQL- and PostgreSQL-compatible relational database that is built for the cloud. It combines the performance and availability of high-end commercial databases with the simplicity and cost-effectiveness of open-source databases. Using Amazon Aurora can reduce your database costs while improving the reliability and availability of the database. As a fully managed service, Aurora is designed to automate time-consuming tasks like provisioning, patching, backup, recovery, failure detection, and repair.

ACTIVITY LOG FOR THE NINE and TENWEEK

Day&Date	Brief Description of the daily activity	Learning Outcome	Person Incharge Signature
DAY 1 17/06/2024	AWS Well - Architected Framework	AWS Well - Architected Framework	
DAY 2 18/06/2024	Reliability and high availability	Reliability and high availability	
DAY 3 19/06/2024	AWS Trusted Advisor	AWS Trusted Advisor	
DAY 4 20/06/2024	Elastic Load Balancing	Elastic Load Balancing	
DAY 5 22/06/2024	Amazon CloudWatch	Amazon CloudWatch	
DAY 6 23/06/2024	Amazon EC2 Auto Scaling	Amazon EC2 Auto Scaling	

WEEKLY REPORT

WEEK-9&10 (From 17/06/2024 to 23/06/2024)

The objective of the activity done: The main objective of this week activities is to Clarify the doubts and increasing the soft skills.

Detailed Report:

Day-1:

Architecture is the art and science of designing and building large structures. Large systems require architects to manage their size and complexity.

Cloud architects:

- Engage with decision makers to identify the business goal and the capabilities that need improvement.
- Ensure alignment between technology deliverables of a solution and the business goals.
- Work with delivery teams that are implementing the solution to ensure that the technology features are appropriate.

Having well-architected systems greatly increases the likelihood of business success.

Day-2:

In the words of Werner Vogels, Amazon's CTO, —Everything fails, all the time. One of the best practices that is identified in the AWS Well-Architected Framework is to plan for failure (or application or workload downtime). One way to do that is to architect your applications and workloads to withstand failure. There are two important factors that cloud architects consider when designing architectures to withstand failure: reliability and availability.

Day-3:

As you have learned so far, you can use the AWS Well-Architected Framework as you design your architectures to understand potential risks in your architecture, identify areas that need improvement, and drive architectural decisions. In this section, you will learn about AWS Trusted Advisor, which is a tool that you can use to review your AWS environment as soon as you start implementing your architectures.

Day-4:

Modern high-traffic websites must serve hundreds of thousands—if not millions—of concurrent requests from users or clients, and then return the correct text, images, video, or application data in a fast and reliable manner. Additional servers are generally required to meet these high volumes. Elastic Load Balancing is an AWS service that distributes incoming application or network traffic across multiple targets—such as Amazon Elastic Compute Cloud (Amazon EC2) instances, containers, internet protocol (IP) addresses, and Lambda functions—in a single Availability Zone or across multiple Availability Zones. Elastic Load Balancing scales

your load balancer as traffic to your application changes over time. It can automatically scale to most workloads.

Day-5:

Amazon CloudWatch is a monitoring and observability service that is built for DevOps engineers, developers, site reliability engineers (SRE), and IT managers. CloudWatch monitors your AWS resources (and the applications that you run on AWS) in real time. You can use CloudWatch to collect and track metrics, which are variables that you can measure for your resources and applications.

Day-6:

When you run your applications on AWS, you want to ensure that your architecture can scale to handle changes in demand. In this section, you will learn how to automatically scale your EC2 instances with Amazon EC2 Auto Scaling.

CONCLUSION:

Claim your badge

- You will receive an email from **Amazon Web Services Training and Certification via Credly** <admin@youracclaim.com> **with a link to claim your digital badge within 24 hours.**
- Click the link in the email and follow the instructions.
- Then share your badge on your LinkedIn or other social media profile to let peers and potential employers know about your accomplishment.
- Once you accept your badge, you can also [download a certificate of completion Links to an external site.](#)
- If you have any questions about your badge, please contact [Credly support Links to an external site. Links to an external site.](#)

AWS Academy is excited to welcome you into our growing global community of students, graduates, and lifelong learners of cloud computing.

Student Self Evaluation of the Short-Term Internship

Student Name:P.Sasi Kumar

Regd No:228A1A05C2

Term of Internship: 2 Months

From: MAY 2024

To: JUNE 2024

Date of Evaluation:

Organization Name & Address: RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS

Please rate your performance in the following areas:

Rating Scale:

Letter grade of CGPA calculation to be provided

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	OVERALL PERFORMANCE	1	2	3	4	5

Date:

Signature of the Student

Evaluation by the Supervisor of the Intern Organization

Student Name: P.V.Sasi

Registration No:228A1A05C2

Term of Internship:

From:

Date of Evaluation

To

Please rate the student's performance in the following areas:

Please note that your evaluation shall be done independent of the Student's self

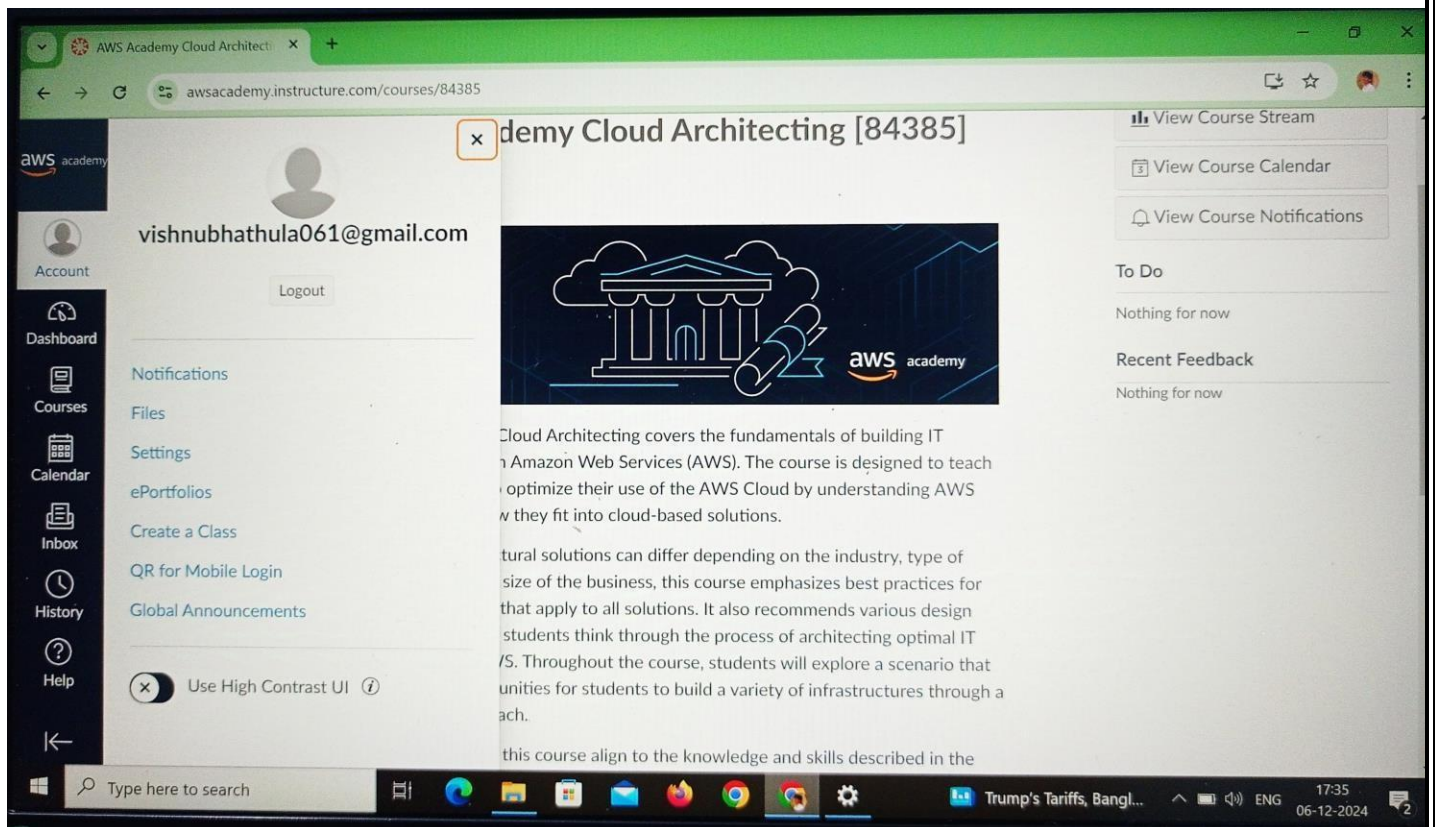
evaluationRating Scale: 1 is lowest and 5 is highest rank

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
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14	Achievement of Desired Outcomes	1	2	3	4	5
15	OVERALL PERFORMANCE	1	2	3	4	5

Date:

Signature of the Supervisor

PHOTOS&VIDEO LINKS



The screenshot displays the AWS Academy Cloud Architecting [84385] course page. The user is logged in as vishnubhathula061@gmail.com. The course description states: "Cloud Architecting covers the fundamentals of building IT on Amazon Web Services (AWS). The course is designed to teach students how to optimize their use of the AWS Cloud by understanding AWS services and how they fit into cloud-based solutions. Architectural solutions can differ depending on the industry, type of size of the business, this course emphasizes best practices for that apply to all solutions. It also recommends various design patterns that students think through the process of architecting optimal IT solutions. Throughout the course, students will explore a scenario that unifies for students to build a variety of infrastructures through a hands-on approach. This course aligns to the knowledge and skills described in the AWS Cloud Practitioner Essentials course."

The sidebar on the left includes the following navigation options: Account, Dashboard, Courses, Calendar, Inbox, History, Help, and a search bar. The right sidebar contains links to View Course Stream, View Course Calendar, View Course Notifications, To Do (Nothing for now), and Recent Feedback (Nothing for now).



ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A Statutory Body of the Government of Andhra Pradesh)

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